

FUNDAMENTALS OF STRENGTH OF MATERIALS
PROF. D.S. PAOLINO – G. BATTIATO – C. BOURSIER
SIMULATION OF TEST 1

Surname:	Name:
ID (matricola):	

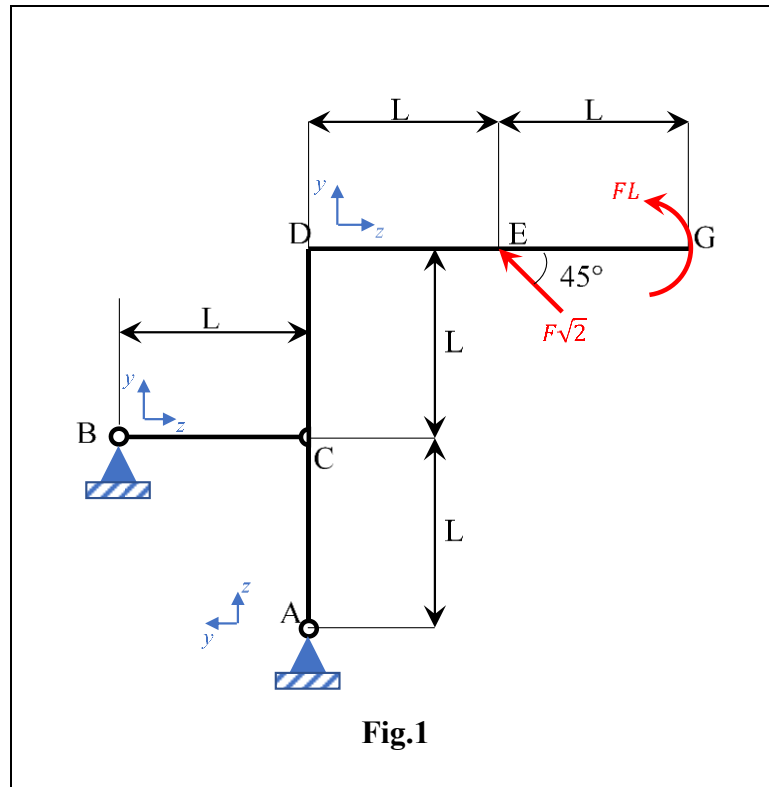
Write here the First number $P = \dots\dots$ and the Last number: $Q = \dots\dots$
of your *ID* (“matricola” on Your University card) to be used to solve the exercises
(example: $ID = 667890$: $P = 6$ and $Q = 0$)

Rules:

1. Exam time is about **2 hours**.
2. **Write very clearly the solutions** of the proposed exercises: if any lack of interpretation occurs, score will be decreased.
3. Written test is sufficient only if score is at least **18/30** and **only in this case the student can attend the second test**.
4. Student has only **15 minutes to leave this exam, after this time he/she must remain until the end and he/she must deliver all the material of the test**.
5. Solution can be only **written by means of pens with undeletable ink (no pencils) and never in red**.
6. Student **can use**: textbooks, lecture notes, didactic material of tutorials.
7. Student **cannot use**: **PORTABLE, LAPTOP, PROGRAMMABLES, IPHONE, IPOD, IPAD...** or any other electronic device that can be connected to other systems or to the network.
8. In case the Student is found working on those devices switched on, **he/she will be expelled**.
9. **Communication among students is forbidden** and in case anybody is found communicating, **he/she will be expelled**.
10. To attend the exam, each student has to show a **valid Identity Document** with a **clear picture** which could definitely certify the correspondence between the candidate and the document. In absence of this evidence the candidate will be rejected and the attendance to test shall be forbidden.
11. Each student is asked **to book the exam** to demonstrate that his/her attendance is allowed and regular. In case some do not book the exam, they will be accepted, but **if attendance will be found irregular, score of the exam will be ANNULLED** and they will repeat the exam, without any exception.
12. In case **the student attends both the written and oral tests and the attendance is then found irregular** by the Administration of Politecnico, he/she **will lose the related score** and will repeat the whole exam.

During the written test, NO QUESTIONS are allowed, being all the relevant inputs written in this document
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EX #1 (16 points)



The structure depicted in Fig.1 is made of **two** beams (**ADG** and **BC**), connected in **C** by an internal hinge. The structure is constrained to ground in **A** and in **B**.

Each beam is made of steel ($S_U=900$ MPa, $S_Y=700$ MPa, $E=2.1\cdot 10^5$ MPa, $\nu=0.29$) and has a hollow circular cross section with outer diameter $D=30$ mm and inner diameter $d=15$ mm.

The 1st body (**ADG**) is loaded by a concentrated force $F\sqrt{2}$ in **E** and by a concentrated moment FL in **G**, being $F = 20 \left(1 + \frac{P-Q}{100}\right)$ kN and $L = 600$ mm.

The local reference frames to be used to describe the stress resultants are depicted for each portion of the two beams. For the computation, the weight of the structure must be neglected.

Questions:

1. Compute the **degree of determinacy** h (1 point).
2. Compute the **external and the internal reactions** (4 points).
3. Compute the **internal actions** N , T_y , M_x (4 points).
4. **Plot the diagrams** of N , T_y , M_x , writing their numerical values at the relevant points (3 points).
5. Compute the **safety factor against buckling** for the beam **BC** (4 points).

EX #2 (14 points)

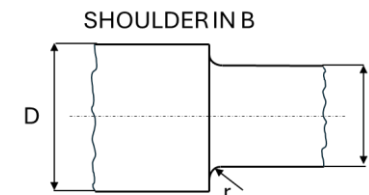
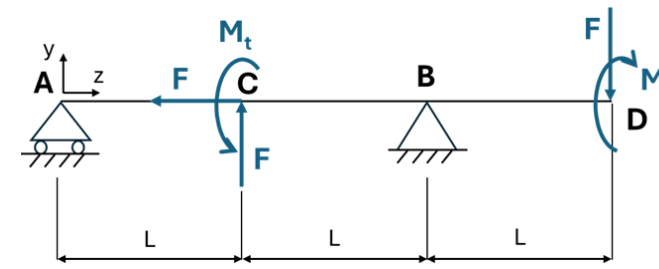


Fig. 2 shows the loads acting on a shaft, constrained to the ground with a pin in A and a hinge in B. In addition to the force $F = 5$ kN, in points C and D, torsional moments $M_t = 100$ Nm are applied. Consider $L = 100$ mm.

The shaft is made of steel ($S_Y=450$ MPa, $E=2.1\cdot 10^5$ MPa, $\nu=0.29$) and has a circular solid cross section with diameter $D=30$ mm. In B a shoulder is present as shown in Fig. 3, with $d=25$ mm and $r=3$ mm.

Questions:

1. Compute the **degree of determinacy** h (1 point).
2. Compute the **external reactions and the internal actions** N , T_y , M_x (4 points).
3. **Plot the diagrams** of N , T_y , M_x , writing their numerical values at the relevant points (3 points).
4. Compute the **safety factor against initial yielding** (3 points).
5. Compute the **safety factor against complete yielding** (3 points).